

CYCLONIC FURNACE

ECO-FRIENDLY SUSPENDED BIOMASS BURNER.



OPERATING PRINCIPLE

The biomass ("rice husk") is conveyed from the storage area by a screw conveyor, whose speed is controlled by an electronic variable-frequency drive (VFD). It is then discharged into a Venturi pneumatic conveying system, which transports it to the combustion chamber, where a rotating cyclonic vortex is generated.

The rice husk burns while suspended in the air, and the gases generated by the combustion spiral are captured by the vortex and directed to the upper combustion chamber, where they are trapped by a second vortex, ensuring complete combustion of the rice husk. The distributor, or mixing chamber, located at the top of the furnace, combines the hot combustion air with the preheated air used to cool the furnace body. The resulting hot air is then drawn by the dryer fan and utilized in the drying process.

After combustion, the resulting ash falls to the bottom of the furnace, forming an ash bed that helps maintain a stable flame. The ash is discharged through the openings in the grate and collected in the ash hopper, which feeds a screw conveyor. This ash-sweeping mechanism operates with a slow, continuous, and intermittent motion, preventing rice husk accumulation and blockage of the discharge grate. The ash exits the furnace at an approximate temperature between 38°C and 45°C, facilitating its handling and packaging.



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Bucaramanga - Colombia

Grain Drying Equipment



- 1 Cyclonic Furnace with Emil Holding Bin HBS-2CL.19 1
- 2 Cyclonic Furnace with Mega Sonora Tower Dryer HBS-4CL.19
- 3 Cyclonic Furnace with GYC Tower Dryer HBS-4CL.24 (2)
- 4 Cyclonic Furnace with Dicorp Holding Bins HBS-4CL.19
- 5 Cyclonic Furnace with ORF PORE Dryer HBS-4CL.19
- 6 Cyclonic Furnace with ORF Double Dryer HBS-4CL.19

ECO-FRIENDLY BIOMASS BURNER

Low emissions of particulate pollutants PCDD and PCDF (commonly known as FURANS and DIOXINS) due to:

- The biomass is burned in suspension while rotating in a spiral motion around the furnace body.
- The combustion of the rice husk biomass entering the combustion chamber is highly efficient, as extremely high combustion temperatures are achieved through the friction of the biomass against the furnace's refractory-lined walls.
- Compared to traditional masonry furnaces or older-generation indirect-fire furnaces, the Cyclonic Furnace provides an effective solution for minimizing environmental pollution.

ADVANTAGES:

- Environmentally Friendly: These units comply with environmental regulations and operate free of smoke and ash emissions. Thanks to the cyclonic (volute) effect, combustion gases are completely burned, preventing contaminating gases and ash from reaching the dryers. As a result, the service life of the dryers' metallic components is significantly extended.
- Proprietary Design: This equipment was developed by EQUIM with the support of British biomass expert Nicholas Neal. It is an original design and not a copy of another manufacturer's technology.
- Proven Reliability: Backed by more than 40 furnace installations operating in rice mills throughout Colombia.
- Low Emissions: Pollutant emissions remain within the limits established by environmental authorities.
- Higher Efficiency: Greater thermal efficiency compared to conventional rice husk furnaces.
- Automatic Combustion Control: Automated system for controlling rice husk combustion, allowing different combustion settings as required.
- Automatic Fuel Feeding System: Fully automated rice husk feeding process.
- Low Power Consumption.
- Low Maintenance and Operating Costs.
- Long Service Life of Refractory Bricks.
- Free of Contaminants that May Damage Materials.
- Guaranteed Quality Materials.
- Compact Design with Minimal Space Requirements.
- Easy Maintenance.
- Long Equipment Service Life.

GENERAL FEATURES

- Mounted on a rigid structural frame.
- The combustion chamber is lined with high-quality refractory brick, capable of withstanding the high temperatures generated during operation.
- At the bottom of the combustion chamber, there is a cast-iron grate equipped with an ash-sweeping mechanism that operates in synchronization with the rice husk feeding system. See **Figure 1**.
- Rice husk feed rate is regulated through a PLC-controlled signal that adjusts the speed of the screw conveyor according to the furnace's combustion requirements.
- The rice husk is supplied to the furnace through a pneumatic conveying system.
- A storage hopper is provided for rice husk storage, with sufficient capacity to supply approximately 2½ hours of continuous operation. See **Figure 2**.
- The mixing chamber (distributor) allows the hot air leaving the combustion chamber to be blended with the cooling air circulating around the furnace body.
- The upper section of the furnace is equipped with a damper that allows fresh air to be introduced at the furnace outlet through the heat transfer duct.
- The upper distribution manifold can accommodate up to four hot-air outlets, depending on the application requirements. See **Figure 3**.

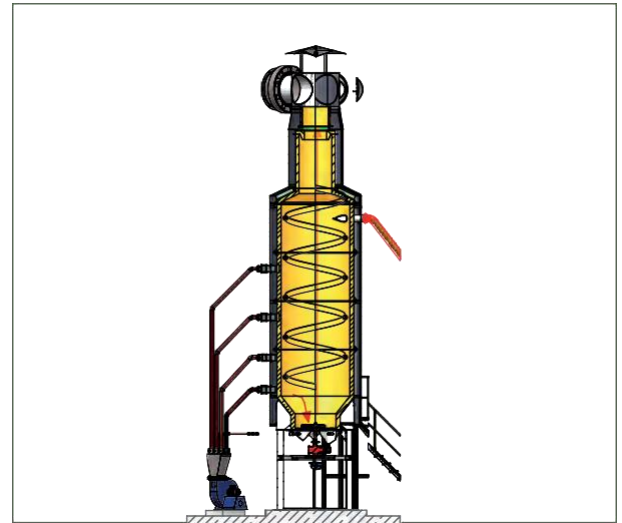


Figura 1



Figura 2



Figura 3

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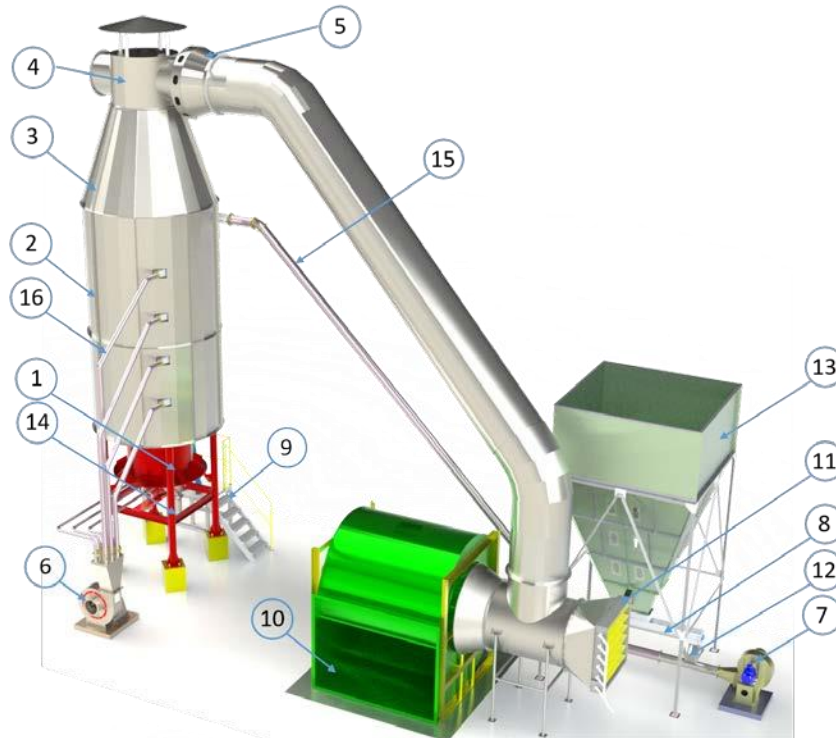
Grain Drying Equipment

TECHNICAL DATA

MODEL	No OF CHAMBERS	HEATING CAPACITY (BTU/h)	POWER (HP)
HBS-2CL.08	2	1.500.000	12.5
HBS-2CL.19	2	4.700.000	12.5
HBS-3CL.19	3	5.000.000	14
HBS-4CL.19	4	7.600.000	14
HBS-4CL.22	4	13.000.000	19
HBS-4CL.24	4	16.000.000	19

Note: PCI Rice husk Lower Heating Value: 12.000 BTU/kg

COMPONENT IDENTIFICATION



ITEM	DESCRIPTION	ITEM	DESCRIPTION
1	HBS-CL Furnace Support Structure	9	HBS-CL Access Platform
2	HBS-CL Furnace Body	10	Ø1500 mm Fan
3	HBS-CL Upper Furnace Body	11	Air Mixing Chamber
4	HBS-CL Hot Air Outlet Duct	12	6-inch Venturi
5	Air Mixing Damper	13	Rice Husk Hopper
6	VR-48 Radial Fan	14	TS-6 Screw Conveyor – Ash Discharge System
7	TB-36.19 Turbine	15	Venturi Coupling Pipe
8	TS-6 Screw Conveyor – Rice Husk Feed System	16	Secondary Air Piping

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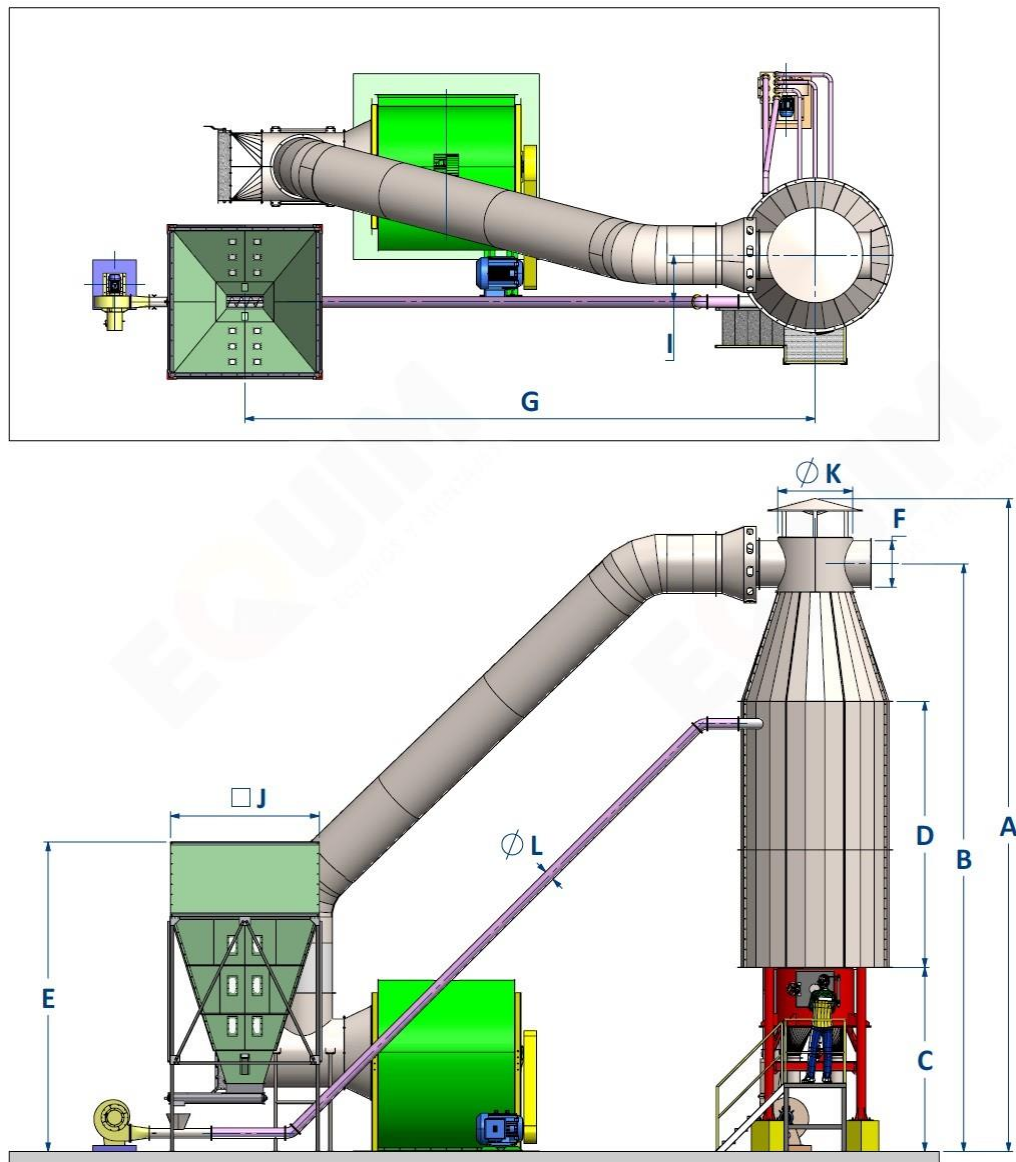
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OVERALL DIMENSIONS



No CHAMBER	MODEL	OVERALL DIMENSIONS											
		A	B	C	D	E	F	G	H	I	J	K	L
2	HBS-2CL.08	4.32	4.0	1.05	2.00	4.80	0,50	4.30	1.25	0.30	2.4	0.70	0.09
2	HBS-2CL.19	9.40	8.30	2.76	3.10	4.80	0.95	7.19	2.30	0.78	2.4	1.20	0.17
3	HBS-3CL.19	10.0	8.90	2.76	3.7	4.80	0.95	7.19	2.30	0.78	2.4	1.20	0.17
4	HBS-4CL.19	10.6	9.5	2.76	4.30	4.80	0.95	7.19	2.30	0.78	2.4	1.20	0.17
4	HBS-4CL.22	11.5	10.3	1.95	5.54	4.95	0.95	7.2	2.5	0.87	2.4	1.30	0.21
4	HBS-4CL.24	11.8	10.6	1.95	5.54	4.95	1.05	7.29	2.75	0.97	2.4	1.40	0.21